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2
3 47562028
4
5 Cohort A
6
7 #Question 1
8 #Create Database
9 create database moviedb;
10
11 #Use database
12 use moviedb;
13 Result:      #    Time      Action  Message Duration / Fetch
14 3    1    11:59:14      use moviedb 0 row(s) affected    0.016 sec
15
16 #Create Tables
17 #Membership Table
18 create table MEMBERSHIP(
19     MEM_NUM int,
20     MEM_FNAME varchar(20),
21     MEM_LNAME varchar(20),
22     MEM_STREET varchar(30),
23     MEM_CITY varchar(50),
24     MEM_STATE varchar(30),
25     MEM_ZIP varchar(20),
26     MEM_BALANCE double,
27     primary key(MEM_NUM)
28
29 );
30
31 describe membership;
32
33 Result:
34 MEM_NUM int NO PRI
35 MEM_FNAME varchar(20) YES
36 MEM_LNAME varchar(20) YES
37 MEM_STREET varchar(30) YES
38 MEM_CITY varchar(50) YES
39 MEM_STATE varchar(30) YES
40 MEM_ZIP varchar(20) YES
41 MEM_BALANCE double YES
42
43
44 #Rental Table
45
46 create table RENTAL(
47     MEMBER_NUM int,
48     RENT_NUM int,
49     RENT_DATE date,
50     foreign key(MEMBER_NUM) references MEMBERSHIP(MEM_NUM),
51     primary key(RENT_NUM)
52
53 );
54
55 describe rental;
56 Result:
57
58 MEMBER_NUM int YES MUL
59 RENT_NUM int NO PRI
60 RENT_DATE date YES
61
62
63 #Price Table
64 create table price (
65     PRICE_CODE int,
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66     PRICE_DESCRIPTION varchar(50),
67     PRICE_RENTFEE double,
68     PRICE_DAILYLATEFEE double,
69     primary key(PRICE_CODE)
70 );
71
72 describe price;
73 Result:
74 PRICE_CODE  int NO  PRI
75 PRICE_DESCRIPTION  varchar(50) YES
76 PRICE_RENTFEE  double YES
77 PRICE_DAILYLATEFEE  double YES
78
79 #Movie Table
80 create table movie(
81     MOVIE_NUM int,
82     MOVIE_TITLE varchar(30),
83     MOVIE_YEAR int,
84     MOVIE_COST double,
85     MOVIE_GENRE varchar(25),
86     MOVIE_PRICE int,
87     primary key(MOVIE_NUM),
88     foreign key(MOVIE_PRICE) references price(PRICE_CODE)
89 );
90 describe movie;
91 Result:
92 MOVIE_NUM  int NO  PRI
93 MOVIE_TITLE  varchar(30) YES
94 MOVIE_YEAR  int YES
95 MOVIE_COST  double YES
96 MOVIE_GENRE  varchar(25) YES
97 MOVIE_PRICE  int YES MUL
98
99 #Video Table
100 create table video(
101     MOVIE_NUMBER int,
102     VID_NUM int,
103     VID_INDATE date,
104     primary key(VID_NUM),
105     foreign key(MOVIE_NUMBER) references movie(MOVIE_NUM)
106 );
107 describe video;
108 Result:
109 MOVIE_NUMBER  int YES MUL
110 VID_NUM int NO  PRI
111 VID_INDATE  date  YES
112
113
114 #DetailRental
115 create table DetailRental  (
116     RENT_NUMBER int,
117     VIDEO_NUMBER int,
118     DETAIL_FEE double,
119     DETAIL_DUEDATE date,
120     DETAIL_RETURNDATE date,
121     DETAIL_DAILYLATEFEE double,
122     primary key(RENT_NUMBER,VIDEO_NUMBER)
123     foreign key(RENT_NUMBER) references rental(RENT_NUM) ,
124     foreign key(VIDEO_NUMBER) references video(VID_NUM)
125 );
126
127 describe DetailRental;
128
129 Result:
130 RENT_NUMBER int NO  PRI

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131 VIDEO_NUMBER      int NO PRI
132 DETAIL_FEE        double YES
133 DETAIL_DUEDATE     date      YES
134 DETAIL_RETURNDATE  date      YES
135 DETAIL_DAILYLATEFEE double YES
136
137
138
139
140 #Question2
141 #Inserting into membership
142 insert into membership
143 values
144 (102,"Tami","Dawson",'2632 Takli Circle',"Norene","TN","37136",11),
145 (103,"Curt","Knight","4025 Cornell Court ","Flatgap","KY ","41219 ",6),
146 (104, "Jamal ","Melendez"," 788 East 145th Avenue"," Quebeck ","TN","38579 ",0) ,
147 (105 , "Iva"," Mcclain"," 6045 Musket Ball Circle ","Summit"," KY ","42783 ",15 ),
148 (106," Miranda"," Parks ","4469 Maxwell Place"," Germantown"," TN","38183", 0 ),
149 (107," Rosario ","Elliott ","7578 Danner Avenue"," Columbia"," TN","38402", 5 ),
150 (108," Mattie"," Guy"," 4390 Evergreen Street"," Lily ","KY ","40740", 0 ),
151 (109," Clint ","Ochoa ","1711 Elm Street ","Greeneville"," TN ","37745 ",10 ),
152 (110," Lewis ","Rosales"," 4524 Southwind Circle"," Counce ","TN","38326", 0 ),
153 (111," Stacy ","Mann"," 2789 East Cook Avenue ","Murfreesboro"," TN","37132", 8 ),
154 (112," Luis"," Trujillo"," 7267 Melvin Avenue ","Heiskell ","TN","37754", 3 ),
155 (113 ,"Minnie ","Gonzales ","6430 Vasili Drive"," Williston"," TN","38076", 0 );
156
157
158 select * from membership;
159 Result:
160 102 Tami      Dawson  2632 Takli Circle   Norene  TN  37136   11
161 103 Curt      Knight  4025 Cornell Court  Flatgap KY   41219    6
162 104 Jamal      Melendez    788 East 145th Avenue  Quebeck  TN    38579   0
163 105 Iva  Mcclain    6045 Musket Ball Circle   Summit   KY    42783   15
164 106 Miranda    Parks    4469 Maxwell Place   Germantown TN  38183   0
165 107 Rosario    Elliott    7578 Danner Avenue   Columbia  TN  38402   5
166 108 Mattie  Guy    4390 Evergreen Street   Lily     KY  40740   0
167 109 Clint  Ochoa    1711 Elm Street      Greeneville TN    37745   10
168 110 Lewis  Rosales  4524 Southwind Circle   Counce   TN    38326   0
169 111 Stacy  Mann    2789 East Cook Avenue  Murfreesboro  TN  37132   8
170 112 Luis   Trujillo  7267 Melvin Avenue     Heiskell  TN    37754   3
171 113 Minnie  Gonzales  6430 Vasili Drive      Williston  TN  38076   0
172
173
174 #inserting into rental
175 insert into rental
176 values
177 (1001,'2016-03-01',103 ),
178 (1002,'2016-03-01',105 ),
179 (1003,'2016-03-02',102 ),
180 (1004,'2016-03-02',110 ),
181 (1005,'2016-03-02',111 ),
182 (1006,'2016-03-02',107 ),
183 (1007,'2016-03-02',104 ),
184 (1008,'2016-03-03',105 ),
185 (1009,'2016-03-03',111 );
186
187 select * from rental;
188 Result:
189 103 1001      2016-03-01
190 105 1002      2016-03-01
191 102 1003      2016-03-02
192 110 1004      2016-03-02
193 111 1005      2016-03-02
194 107 1006      2016-03-02
195 104 1007      2016-03-02

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196 105 1008      2016-03-03
197 111 1009      2016-03-03
198
199
200 #inserting into price
201 insert into price
202 values
203 (1,"Standard",2 ,1 ),
204 (2,"New Release", 3.5 ,3 ),
205 (3,"Discount",1.5, 1 ),
206 (4 ,"Weekly Special", 1, .5 );
207
208 select * from price;
209 Result:
210 1   Standard      2      1
211 2   New Release   3.5    3
212 3   Discount      1.5    1
213 4   Weekly Special 1      0.5
214
215 #inserting into movie
216 insert into movie
217 values
218 (1234 ,"The Cesar Family Christmas", 2014, 39.95,"FAMILY",2 ),
219 (1235,"Smokey Mountain Wildlife", 2011, 59.95,"ACTION", 1 ),
220 (1236,"Richard Goodhope",2015 ,59.95 ,"DRAMA", 2 ),
221 (1237,"Beatnik Fever", 2014 ,29.95 ,"COMEDY", 2 ),
222 (1238,"Constant Companion", 2015, 89.95 ,"DRAMA",null ),
223 (1239,"Where Hope Dies", 2005 ,25.49 ,"DRAMA", 3 ),
224 (1245,"Time to Burn",2012,45.49,"ACTION", 1 ),
225 (1246,"What He Doesn't Know",2013 ,58.29,"COMEDY", 1 );
226
227 select * from movie;
228 Result:
229 1234   The Cesar Family Christmas   2014      39.95   FAMILY   2
230 1235    Smokey Mountain Wildlife     2011      59.95   ACTION    1
231 1236    Richard Goodhope              2015      59.95   DRAMA     2
232 1237    Beatnik Fever                  2014      29.95   COMEDY    2
233 1238    Constant Companion             2015      89.95   DRAMA     null
234 1239    Where Hope Dies                2005      25.49   DRAMA     3
235 1245    Time to Burn                   2012      45.49   ACTION    1
236 1246    What He Doesn't Know           2013      58.29   COMEDY    1
237
238
239 #inserting into video
240 insert into video
241 values
242 (54321,'2015-06-18',1234),
243 (54324,'2015-06-18',1234),
244 (54325,'2015-06-18',1234),
245 (34341,'2014-01-22',1235),
246 (34342,'2014-01-22',1235),
247 (34366,'2016-03-02',1236),
248 (34367,'2016-03-02',1236),
249 (34368,'2016-03-02',1236),
250 (34369,'2016-03-02',1236),
251 (44392,'2015-10-21',1237),
252 (44397,'2015-10-21',1237),
253 (59237,'2016-02-14',1237),
254 (61388,'2014-01-25',1239),
255 (61353,'2013-01-28',1245),
256 (61354,'2013-01-28',1245),
257 (61367,'2015-07-30',1246),
258 (61369,'2015-07-30',1246);
259
260 select * from video;

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261 Result:
262 1235      34341      2014-01-22
263 1235      34342      2014-01-22
264 1236      34366      2016-03-02
265 1236      34367      2016-03-02
266 1236      34368      2016-03-02
267 1236      34369      2016-03-02
268 1237      44392      2015-10-21
269 1237      44397      2015-10-21
270 1234      54321      2015-06-18
271 1234      54324      2015-06-18
272 1234      54325      2015-06-18
273 1237      59237      2016-02-14
274 1245      61353      2013-01-28
275 1245      61354      2013-01-28
276 1246      61367      2015-07-30
277 1246      61369      2015-07-30
278 1239      61388      2014-01-25
279
280
281
282 #inserting into DetailRental
283 insert into DetailRental
284 values
285 (1001,34342,2,'2016-03-04','2016-03-02',null),
286 (1001,61353,2,'2016-03-04','2016-03-03',1),
287 (1002,59237,3.5,'2016-03-04','2016-03-04',3),
288 (1003,54325,3.5,'2016-03-04','2016-03-09',3),
289 (1003,61369,2,'2016-03-06','2016-03-09',1),
290 (1003,61388,0,'2016-03-06','2016-03-09',1),
291 (1004,44392,3.5,'2016-03-05','2016-03-07',3),
292 (1004,34367,3.5,'2016-03-05','2016-03-07',3),
293 (1004,34341,2,'2016-03-07','2016-03-07',1),
294 (1005,34342,2,'2016-03-07','2016-03-05',1),
295 (1005,44397,3.5,'2016-03-05','2016-03-05',3),
296 (1006,34366,3.5,'2016-03-05','2016-03-04',3),
297 (1006,61367,2,'2016-03-07',null,1),
298 (1007,34368,3.5,'2016-03-05',null,3),
299 (1008,34369,3.5,'2016-03-05','2016-03-05',3),
300 (1009,54324,3.5,'2016-03-05',null,3),
301 (1001,34366,3.5,'2016-03-04','2016-03-02',3);
302
303 select * from DetailRental;
304
305 Result:
306 1001      34342      2      2016-03-04      2016-03-02
307 1001      61353      2      2016-03-04      2016-03-03      1
308 1002      59237      3.5    2016-03-04      2016-03-04      3
309 1003      54325      3.5    2016-03-04      2016-03-09      3
310 1003      61369      2      2016-03-06      2016-03-09      1
311 1003      61388      0      2016-03-06      2016-03-09      1
312 1004      44392      3.5    2016-03-05      2016-03-07      3
313 1004      34367      3.5    2016-03-05      2016-03-07      3
314 1004      34341      2      2016-03-07      2016-03-07      1
315 1005      34342      2      2016-03-07      2016-03-05      1
316 1005      44397      3.5    2016-03-05      2016-03-05      3
317 1006      34366      3.5    2016-03-05      2016-03-04      3
318 1006      61367      2      2016-03-07                      1
319 1007      34368      3.5    2016-03-05                      3
320 1008      34369      3.5    2016-03-05      2016-03-05      3
321 1009      54324      3.5    2016-03-05                      3
322 1001      34366      3.5    2016-03-04      2016-03-02      3
323
324
325

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326 #Question3
327
328 select movie_year
329 from movie
330 where movie_num = 1245;
331
332 Result Before Change:
333 2012
334
335 update movie
336 set MOVIE_YEAR = 2010
337 where MOVIE_NUM = 1245;
338
339 select movie_year
340 from movie
341 where movie_num=1245;
342
343 Result After Change:
344 2010
345
346
347 #Question4
348
349 select movie_price
350 from movie
351 where movie_genre = "ACTION";
352
353 Result Before Change:
354 1
355 1
356
357 update movie
358 set movie_price =3
359 where movie_genre ="ACTION";
360
361 select movie_price
362 from movie
363 where movie_genre = "ACTION";
364
365 Result After Change:
366 3
367 3
368
369 #Question 5
370 select price_rentfee
371 from price;
372
373
374 Result Before Change:
375 2
376 3.5
377 1.5
378 1
379
380 update price
381 set price_rentfee = price_rentfee+0.50;
382
383 select price_rentfee
384 from price;
385
386 Result After Change:
387 2.5
388 4
389 2
390 1.5
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391
392
393 #Question6
394 select movie_title,movie_year,movie_genre
395 from movie;
396
397 Result:
398 The Cesar Family Christmas    2014    FAMILY
399 Smokey Mountain Wildlife      2011    ACTION
400 Richard Goodhope              2015    DRAMA
401 Beatnik Fever                 2014    COMEDY
402 Constant Companion            2015    DRAMA
403 Where Hope Dies               2005    DRAMA
404 Time to Burn                  2010    ACTION
405 What He Doesn't Know          2013    COMEDY
406
407 #Question7
408
409 select movie_year,movie_title,movie_cost
410 from movie
411 order by movie_year desc;
412
413 Result:
414 2015    Richard Goodhope      59.95
415 2015    Constant Companion    89.95
416 2014    The Cesar Family Christmas  39.95
417 2014    Beatnik Fever        29.95
418 2013    What He Doesn't Know   58.29
419 2011    Smokey Mountain Wildlife  59.95
420 2010    Time to Burn          45.49
421 2005    Where Hope Dies       25.49
422
423 #Question 8
424
425 select movie_title,movie_year,movie_genre
426 from movie
427 order by movie_genre asc,movie_year desc;
428
429 Result:
430 Smokey Mountain Wildlife      2011    ACTION
431 Time to Burn                  2010    ACTION
432 Beatnik Fever                 2014    COMEDY
433 What He Doesn't Know          2013    COMEDY
434 Richard Goodhope              2015    DRAMA
435 Constant Companion            2015    DRAMA
436 Where Hope Dies               2005    DRAMA
437 The Cesar Family Christmas    2014    FAMILY
438
439 #Question9
440
441 select movie_num,movie_title,movie_price
442 from movie
443 where movie_title like 'R%';
444
445 Result:
446 1236    Richard Goodhope      2
447
448 #Question 10
449 select movie_title,movie_year,movie_cost
450 from movie
451 where movie_title like '%hope%'
452 order by movie_title asc;
453
454 Result:
455 Richard Goodhope      2015    59.95

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456 Where Hope Dies      2005      25.49
457
458 #Question 11
459 select movie_title,movie_year,movie_genre
460 from movie
461 where movie_genre = 'ACTION';
462
463 Result:
464 Smokey Mountain Wildlife      2011      ACTION
465 Time to Burn                  2010      ACTION
466
467
468 #Question 12
469 select movie_num,movie_title,movie_cost
470 from movie
471 where movie_cost > 40;
472
473 Result:
474 1235      Smokey Mountain Wildlife      59.95
475 1236      Richard Goodhope              59.95
476 1238      Constant Companion            89.95
477 1245      Time to Burn                  45.49
478 1246      What He Doesn't Know          58.29
479
480
481 #Question 13
482 select movie_num,movie_title,movie_cost,movie_genre
483 from movie
484 where movie_genre in ('ACTION','COMEDY') and movie_cost <50
485 order by movie_genre asc;
486
487
488 Result:
489 1245      Time to Burn      45.49      ACTION
490 1237      Beatnik Fever    29.95      COMEDY
491
492 #Question 14
493 select mem_num,mem_fname,mem_lname,mem_street,mem_state,mem_balance
494 from membership
495 where mem_state = 'TN'
496      and mem_balance < 5
497      and mem_street like '%Avenue';
498
499 Result:
500 104      Jamal      Melendez      788 East 145th Avenue      TN      0
501 112      Luis      Trujillo      7267 Melvin Avenue      TN      3
502
503 #Question 15
504 select movie_genre, count(*) as Number_Of_Movies
505 from movie
506 group by movie_genre;
507
508 Result:
509 FAMILY      1
510 ACTION      2
511 DRAMA      3
512 COMEDY      2
513
514
515 #Question 16
516
517 select round(avg(movie_cost),4) as Average_Movie_Cost
518 from movie;
519
520 Result:

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521 51.1275
522
523 #Question17
524 select movie_genre, round(avg(movie_cost),2) as Average_Cost
525 from movie
526 group by movie_genre;
527
528 Result:
529 FAMILY 39.95
530 ACTION 52.72
531 DRAMA 58.46
532 COMEDY 44.12
533
534
535 #Question 18
536 select movie_title,movie_genre,price_description,price_rentfee
537 from movie
538 join price
539 on movie.movie_price=price.price_code;
540
541 Result:
542 What He Doesn't Know COMEDY Standard 2.5
543 The Cesar Family Christmas FAMILY New Release 4
544 Richard Goodhope DRAMA New Release 4
545 Beatnik Fever COMEDY New Release 4
546 Smokey Mountain Wildlife ACTION Discount 2
547 Where Hope Dies DRAMA Discount 2
548 Time to Burn ACTION Discount 2
549
550
551 #Question19
552 select movie_genre,round(avg(price_rentfee),2) as Average_Rental_Fee
553 from movie
554 join price
555 on movie.movie_price=price.price_code
556 group by movie_genre;
557
558
559 Result:
560 COMEDY 3.25
561 FAMILY 4
562 DRAMA 3
563 ACTION 2
564
565
566 #Question20
567 select movie_title, round((movie_cost/price_rentfee),2) as breakeven_rentals
568 from movie
569 join price
570 on movie.movie_price=price.price_code
571 where movie_price is not null;
572
573 Result:
574 What He Doesn't Know 23.32
575 The Cesar Family Christmas 9.99
576 Richard Goodhope 14.99
577 Beatnik Fever 7.49
578 Smokey Mountain Wildlife 29.98
579 Where Hope Dies 12.74
580 Time to Burn 22.74
581
582 #Question21
583 select movie_title,movie_year
584 from movie
585 where movie_price is not null;

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586
587 Result:
588 What He Doesn't Know      2013
589 The Cesar Family Christmas 2014
590 Richard Goodhope         2015
591 Beatnik Fever            2014
592 Smokey Mountain Wildlife   2011
593 Where Hope Dies          2005
594 Time to Burn             2010
595
596
597 #Question22
598 select movie_title,movie_genre,movie_cost
599 from movie
600 where movie_cost between 44.99 and 49.99;
601
602 Result:
603 Time to Burn      ACTION  45.49
604
605 #Question23
606 select movie_title,price_description,price_rentfee,movie_genre
607 from movie
608 join price
609 on movie.movie_price=price.price_code
610 where movie_genre in ('Family','Comedy','Drama');
611
612 Result:
613 The Cesar Family Christmas  New Release 4   FAMILY
614 Richard Goodhope          New Release 4   DRAMA
615 Beatnik Fever             New Release 4   COMEDY
616 Where Hope Dies           Discount    2   DRAMA
617 What He Doesn't Know      Standard    2.5 COMEDY
618
619
620 #Question24
621 select distinct  mem_num,mem_fname,mem_lname,mem_balance
622 from membership
623 join rental
624 on membership.mem_num=rental.member_num;
625
626
627 Result:
628 102 Tami      Dawson  11
629 103 Curt      Knight   6
630 104 Jamal     Melendez   0
631 105 Iva       Mcclain   15
632 107 Rosario   Elliott    5
633 110 Lewis     Rosales   0
634 111 Stacy     Mann     8
635
636 #Question25
637 select min(mem_balance) as Minimum_Balance,max(mem_balance) as
Maximum_Balance,round(avg(mem_balance),2) as Average_Balance
638 from membership
639 join rental
640 on membership.mem_num=rental.member_num;
641
642 Result:
643 0    15  7.56
644
645 #Question26
646 select rent_number,rent_date,video_number,movie_title,detail_duedate,detail_returndate
647 from DetailRental
648 join rental
649 on rental.rent_num = DetailRental.rent_number

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650 join video
651 on video.vid_num = DetailRental.video_number
652 join movie
653 on movie.movie_num = video.movie_number
654 where detail_returndate > detail_duedate
655 order by rent_number ,movie_title;
656
657 Result:
658 1003      2016-03-02   54325   The Cesar Family Christmas   2016-03-04   2016-03-09
659 1003      2016-03-02   61369   What He Doesn't Know        2016-03-06   2016-03-09
660 1003      2016-03-02   61388   Where Hope Dies             2016-03-06   2016-03-09
661 1004      2016-03-02   44392   Beatnik Fever               2016-03-05   2016-03-07
662 1004      2016-03-02   34367   Richard Goodhope            2016-03-05   2016-03-07
663
664 #Question27
665 select rent_number,rent_date,movie_title,detail_fee
666 from DetailRental
667 join rental
668 on rental.rent_num = DetailRental.rent_number
669 join video
670 on video.vid_num = DetailRental.video_number
671 join movie
672 on movie.movie_num = video.movie_number
673 where detail_returndate <= detail_duedate
674 order by movie_title;
675
676 Result:
677 1002      2016-03-01   Beatnik Fever      3.5
678 1005      2016-03-02   Beatnik Fever      3.5
679 1006      2016-03-02   Richard Goodhope   3.5
680 1008      2016-03-03   Richard Goodhope   3.5
681 1001      2016-03-01   Richard Goodhope   3.5
682 1001      2016-03-01   Smokey Mountain Wildlife  2
683 1004      2016-03-02   Smokey Mountain Wildlife  2
684 1005      2016-03-02   Smokey Mountain Wildlife  2
685 1001      2016-03-01   Time to Burn       2
686
687 #Question28
688 select movie_num, movie_genre, movie_cost,
689        round(avg(movie_cost) over (partition by movie_genre),2) as Average_Cost,
690        round(((movie_cost - avg(movie_cost) over (partition by movie_genre)) /
691        avg(movie_cost) over (partition by movie_genre)) * 100, 2) as Percent_Difference
692 from movie
693 order by movie_genre, movie_num;
694
695 Result:
696 1235      ACTION    59.95    52.72    13.71
697 1245      ACTION    45.49    52.72   -13.71
698 1237      COMEDY    29.95    44.12   -32.12
699 1246      COMEDY    58.29    44.12    32.12
700 1236      DRAMA     59.95    58.46     2.54
701 1238      DRAMA     89.95    58.46    53.86
702 1239      DRAMA     25.49    58.46   -56.4
703 1234      FAMILY    39.95    39.95     0

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